

1 User Authentication Flow

Secure Login Gateway

- User enters username and password on the login screen.
- Password is hashed using SHA-256 via the Web Crypto API.
- Hash is validated against stored admin credential hashes.
- Username matching is case-insensitive with whitespace trimmed.

Validate

Session Initialization

- User role and identity are stored in the browser session.
- Workspace unlock flag is activated for dashboard access.
- All users land on the Projects Registry as the default tab.
- Admin profile dropdown is configured per authenticated user.

2 User Roles & Permission Matrix

Super Admin

Shahroz

Owner

Diana, Greg, Monica

Accountant

Colleen

Shift Lead

Donna

Field QRE

Hugo, Nabil, Rogelio, Clarence

Customer

AutoKabel, Magna, Hutchinson, Brose

Feature / Tab	Super Admin	Owner	Accountant	Lead	QRE	Customer
Incident Defects Feed	✓	✓	-	✓	-	-
Visual Defect Heatmap	✓	✓	-	✓	-	-
Daily Tasks Planner	✓	✓	-	✓	-	-
Shift Summaries & Publish	✓	✓	-	✓	-	Published
Suppliers Directory	✓	✓	-	-	-	-
Timesheets & Mileage	✓	✓	✓	-	Own Only	Own Client
Invoice & Billing	✓	✓	✓	-	-	-
Rework Logs Feed	✓	✓	-	✓	-	-
Email Audit Logs	✓	✓	-	-	-	-
User Directory	✓	✓	-	-	-	-
Projects Registry	✓	✓	-	-	-	-
Setup & Onboarding	✓	✓	-	-	-	-
System Events Logger	✓	-	-	-	-	-
Launch Roadmap	✓	-	-	-	-	-
Customer Portal	-	-	-	-	-	✓
Pulse AI Chatbot	✓	✓	✓	✓	✓	-

3 Profile Switching & Session Security

Admin Profile Dropdown

- Admins can switch views: Greg (Owner), Colleen (Finance), Donna (Lead).
- Super Admin option is only visible when logged in as Shahroz.
- Diana and other users are blocked from accessing the Super Admin profile.

Session Controls

- Lock Session clears the browser session and forces re-authentication.
- Database Reset wipes all local data, re-seeds defaults, and re-syncs all 15 data collections to Supabase PostgreSQL cloud database.

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Phone Simulator - Complete Field Representative Workflow

Rep Authentication

- Email and password login form.
- Quick login presets for each rep.
- Master passcode bypass available.
- Remember device via local storage.

Auth →

Shift Handover Detection

- Detects if another rep had an active shift.
- Auto-locks the previous rep's draft report.
- Shows handover alert requiring acknowledgment.
- Records handover event in system logs.

ACK →

Home Dashboard

- User profile with avatar, name, and role.
- Plant location selector dropdown.
- Online/offline status indicator.
- Six action buttons and daily task list.

Clock In / Clock Out System

- Clock In creates a draft shift report and records the start time in local storage.
- Clock Out shows elapsed hours calculation (demo mode uses 9.0 hours if under 1 minute).
- Mileage is tracked per time entry through the dashboard mileage input field.

Daily Tasks Viewer

- Tasks filtered by representative and today's date from the cloud database.
- Each task can be toggled between completed and pending status inline.
- Tasks are dispatched by the admin from the Daily Tasks Planner tab on the dashboard.

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Incident Report Wizard - 4 Steps with AI Duplicate Check

1 Capture Photos

Three photo slots: wide, medium, and close-up views. Mock camera with shutter button. Canvas overlay for freehand red annotation drawing. Defect location pin placement on a part template diagram.



2 Barcode Scan

Mock barcode scanner with selectable part numbers from the database. QR code scanner for bin location lookup. Manual entry option with audio warning beep. Multi-part list builder to add, remove, and set quantities.



3 Describe Defect

Area selector and AI-suggested defect types based on scanned part. Free-text description and actions taken. Classification as PRR or QR. Toggle switches for sort required, RMA required, and defect returned.



3.5 AI Dedup Check

Jaccard similarity analysis compares against incidents from the past 24 hours. Displays match percentage. Option to merge into an existing incident or continue as a new report.

Step 4: Review, Preview & Submit

- Full preview of all captured data: photos, annotations, parts, and description.
- Toggle to preview the formatted HTML email notification before sending.
- Two-second sending animation with progress indicator.
- Creates three records: incident report, email delivery log, and system event.

Additional Phone Features

- Rework Logger: select part number, enter quantity sorted, hours spent, and notes.
- Expense Claims: enter amount, category, capture receipt photo, and submit.
- Shift Summary: four-area walk checklist, bonus tasks, and submit final report.
- Audio Feedback: distinct tones for success, scan, and warning events.

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Admin Dashboard - Operations & Quality Management Tabs

Incidents & Defect Feed

- Searchable list of all defect incident reports with supplier and status filters.
- Slide-out detail drawer with photo gallery, annotations, parts list, and coordinates.
- Toggle incident status between Open and Closed (logged to system events).
- Export individual incident as a branded PDF with logo, confidentiality badge, and watermark. Browser print dialog. Resend supplier email notification.

Visual Defect Heatmap Matrix

- Interactive defect location map rendered on a part template image.
- Part number selector dropdown to switch between different components.
- Time scrubber slider to visualize how defect patterns cluster over time.
- Hover tooltips display individual incident details and timestamps.
- Dot opacity is weighted by recency of the defect occurrence.

Daily Tasks Planner

- Date-scoped task list organized by representative assignment.
- Representative filter dropdown to view a specific person's tasks.
- Create new tasks by assigning instruction text, representative, and date.
- Toggle each task between completed and pending status.

Shift Summaries Log

- Chronological list of all shift walkthrough reports submitted by reps.
- Detail drawer showing areas walked, bonus tasks, and incident counts.
- Publish to Customer button changes status, making the report visible in the portal.
- Export as branded shift walkthrough audit PDF or use browser print.

Suppliers Directory

- Grid of supplier cards showing name, invoice schedule, allotted hours, and contacts.
- Each card lists associated plant locations served by that supplier.
- Export full directory as a branded PDF report or use browser print.

Rework Logs Feed

- All rework and sort activities with date-based filtering.
- Detail drawer for each individual rework entry.
- Export as feed-level PDF, individual rework PDF, or browser print.

Email Delivery Logs

- Complete email delivery audit trail for all sent notifications.
- Detail view shows: recipients, CC, subject, HTML body, status, and timestamp.

User Directory

- Grid of all registered user cards: name, email, phone, role, and avatar.
- Displays pay currency assignment for each representative.

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Pulse AI - Conversational Database Auditor

AI Chat Console

- Natural language chat interface with persistent message history.
- Role-specific welcome messages and contextual quick-action chips.
- Admin commands: audit database, scan duplicates, export Excel or CSV.
- Accountant commands: audit timesheets, export financial reports.
- QRE commands: check hours, view assigned tasks for today.

Automated Audit Engine

- Comprehensive scan triggered via chat or the Run Scan button.
- Flags duplicate defect incidents matching on part number and area.
- Detects timesheet anomalies: negative hours, entries exceeding 24 hours.
- Identifies receipt mismatches and checks daily hour limit compliance.
- Directly triggers Excel, CSV, and PDF exports through chat commands.

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Timesheets, Invoicing & Financial Operations

Log Hours & Expenses

- Hours form: representative, supplier, date, hours worked, mileage in km, notes.
- Expense form: representative, supplier, date, category, dollar amount, notes.
- Both records sync to Supabase instantly.

Invoice Generator

- Select supplier and currency (CAD/USD).
- Calculates: hours multiplied by billing rate plus mileage at \$0.73 per km plus expenses.
- Export as branded client invoice PDF.
- Export as QuickBooks-compatible CSV file.

Payroll & Expense Approvals

- Queue of pending expense claims to review.
- View receipt photos in a lightbox modal.
- Approve or reject each expense with status.
- Extra hours admin final approval queue after customer stage. Auto-creates time entry.

Rates Configuration & Admin Onboarding

- Rate override matrix: set custom billing and pay rates per representative and supplier.
- Customers sub-tab: register new client companies with contact name, email, and role.
- Locations sub-tab: map new plant locations, link to supplier, assign representative, set rates.
- Representatives sub-tab: onboard new reps with name, email, phone, and pay currency.

Projects Registry & Quick Add System

- Create project records linking a representative, client, plant, with billing and pay rates.
- Project-level rates take priority over the general rate override matrix.
- Quick Add modals for representatives, clients, and plants from within the registry.
- Each modal writes directly to Supabase and auto-selects the new record in dropdown.

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Extra Hours - Three-Stage Approval Workflow

1 Rep Files Request

Representative submits an overtime request specifying supplier, plant, date, hours, and written justification. Status is set to pending customer approval.



2 Customer Reviews

Customer sees the request in their portal approval queue. They can approve or reject with a written comment. Approved requests escalate to admin review.



3 Admin Approves

Admin reviews the request with customer comment in the Payroll tab. Approval automatically creates a new timesheet entry. Rejection records the reason.



Outcome

- Approved: time entry created with
- tagged note in the timesheets.
- Full audit trail with timestamps
- and comments from each stage.

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Expense Claims - Two-Stage Approval Workflow

1 Rep Submits Claim

Representative files expense via the phone app or dashboard. Includes amount, category, receipt photograph, and notes. Status is set to submitted.



2 Admin Reviews

Admin reviews in the Payroll Approvals panel. Can view receipt photo in a lightbox. Approves or rejects the claim with a status update.



Billing Impact

- Approved expenses are included in client
- invoice calculations and payroll reports.
- Receipt photos are embedded in audit PDFs.

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Customer Portal - Client-Facing Dashboard

Plant Assignment Grid

- Displays the customer's plants with their assigned quality representative.
- Progress bar comparing unbilled vs allotted weekly hours per location.

Overtime Approval Queue

- Customer can approve or reject overtime requests from reps at their plants.
- Comment field for approval rationale.
- Approved requests escalate to admin.

Published Reports & Audit

- View published shift walkthrough reports.
- Weekly hours audit table filtered to the customer's supplier records only.
- Billing rates are hidden from the client.

12 Database Architecture - 15 Entity Collections

Collection Name	Description
users	Representatives, leads, owners, and accountants with profile details and pay currency
rates	Billing and pay rate overrides configured per representative, supplier, and plant
plants	OEM assembly plant locations with addresses and brand associations
suppliers	Client and supplier companies with contact information and invoice schedules
parts	Part number registry used for barcode scanning and defect tracking
incidents	Defect incident reports with annotated photos, parts lists, and coordinates
shiftReports	Shift walkthrough audit summaries submitted by field representatives
reworkLogs	Rework and sort activity logs with quantities, hours, and part numbers
timeEntries	Hours worked and mileage logged per representative per day
emailLogs	Email delivery audit trail for all system-generated notifications
dailyTasks	Daily task assignments dispatched to representatives by administrators
expenseEntries	Expense claims with receipt photographs, amounts, and approval status
projects	Project registry entries linking reps, clients, plants with rates and currency
extraHoursRequests	Overtime approval workflow items with three-stage status tracking
systemLogs	System event audit trail accessible exclusively to the Super Admin

13 Supabase Synchronization - Browser LocalStorage to PostgreSQL

1 Initialize & Pull from Cloud

On application load, the database initializer seeds the local browser cache from default data, then after a brief delay pulls the latest records from Supabase. If the cloud has data, it overwrites local. If the cloud is empty, it seeds Supabase from local defaults.

2 Push on Every Write Operation

Every save operation writes to Browser LocalStorage immediately for zero-latency user experience, then simultaneously fires a background upsert to the Supabase PostgreSQL database. Delete operations call the Supabase delete endpoint by entity ID.

3 Cross-Component Reactivity

After every database write, a custom DOM event is dispatched. Both the Phone Simulator and the Web Dashboard listen for this event and re-read their local state from the updated cache, ensuring both panels stay in sync.

4 Real-Time Audio & Visual Notifications

The dashboard detects new data arrivals via reactive props. It plays a dual-tone synthesizer chime using the Web Audio API and displays a toast notification banner for new incidents, rework logs, shift reports, and expense claims.

5 Role-Based Data Sanitization

The data access layer applies per-role filtering on every read. Representatives see only their own records. Customers see only their supplier's data with billing rates stripped. Administrators receive full unrestricted access to all collections.

14 Complete Report & Export Catalog

PDF Reports - 7 Types

Report Name	Output Filename Pattern	Generated From
Incident Audit Report	IDS_Pulse_Audit_{PartNo}_{Incid}.pdf	Incidents tab - per incident
Shift Walkthrough Report	IDS_Shift_Walkthrough_{Date}_{Srid}.pdf	Shift Logs tab - per report
Supplier Contacts Directory	IDS_Supplier_Contacts_Directory.pdf	Suppliers tab - full directory
Timesheet Payroll Audit	IDS_Timesheets_Audit_Report.pdf	Timesheets tab - full payroll
Rework Feed Report	IDS_Rework_Audit_Feed_{Date}.pdf	Rework Logs tab - all entries
Individual Rework Audit	IDS_Rework_Audit_{Rwld}.pdf	Rework Logs tab - per entry
Client Billing Invoice	Invoice_{ClientName}_{Timestamp}.pdf	Invoice Generator sub-tab

Spreadsheet Exports - 3 Types

Report Name	Output Filename Pattern	Generated From
QuickBooks CSV Export	QuickBooks_Export_{Supplier}.csv	Invoice Generator - per client
Payroll CSV Export	IDS_Timesheets_Payroll_{Date}.csv	Timesheets tab - full payroll
Styled Excel Workbook	IDS_Timesheets_Payroll_{Date}.xlsx	Timesheets tab - via ExcelJS

Browser Print Reports - 5 Types

Report Name	Output Filename Pattern	Generated From
Incident Audit Print	Styled HTML document	Incidents tab - per incident
Shift Walkthrough Print	Styled HTML document	Shift Logs tab - per report
Supplier Directory Print	Styled HTML document	Suppliers tab - full directory
Timesheet Payroll Print	Styled HTML document	Timesheets tab - full payroll
Rework Log Print	Styled HTML document	Rework Logs tab - feed or individual

15 Hidden System Events Logger - Super Admin Exclusive

Tracked Event Categories

- Authentication: login events and session initialization records.
- System: all onboarding operations - create customer, location, representative, save or delete rate overrides, quick-add actions from modals.
- Shift: clock in and out, shift report submissions, publish to customer.
- Incident: new incident creation, status updates between open and closed.
- Rework: rework log submissions.
- Payroll: overtime approvals, expense approvals, CSV, Excel, and PDF exports.

Access Control & Security

- The System Events Logs tab is completely hidden from all non-Super Admin users.
- The sidebar button is wrapped in a role guard that checks for the shahroz role.
- The tab content panel has a secondary guard preventing direct URL navigation.
- Each event records: timestamp, acting user identity, category, action, and details.
- The clear console button resets the log to a single "cleared" audit entry.
- All event records are synchronized to the Supabase systemLogs cloud table.

16 Projects Registry & Quick Add System

Projects Registry Tab

- List of registered projects: rep, client, plant, rates, currency, status.
- Create new project form with billing rate, pay rate, currency per project.
- Projects used by rate resolver - project rates take priority over rate overrides.

Quick Add Modals (Inline Popups)

- Quick Add Rep: name, email, phone, currency - auto-selects in registry.
- Quick Add Client: name, invoice schedule - saves to suppliers collection.
- Quick Add Plant: name, address, linked supplier - saves to plants collection.
- Available from Projects Registry tab and Rates Config sub-tab.